



NCAR
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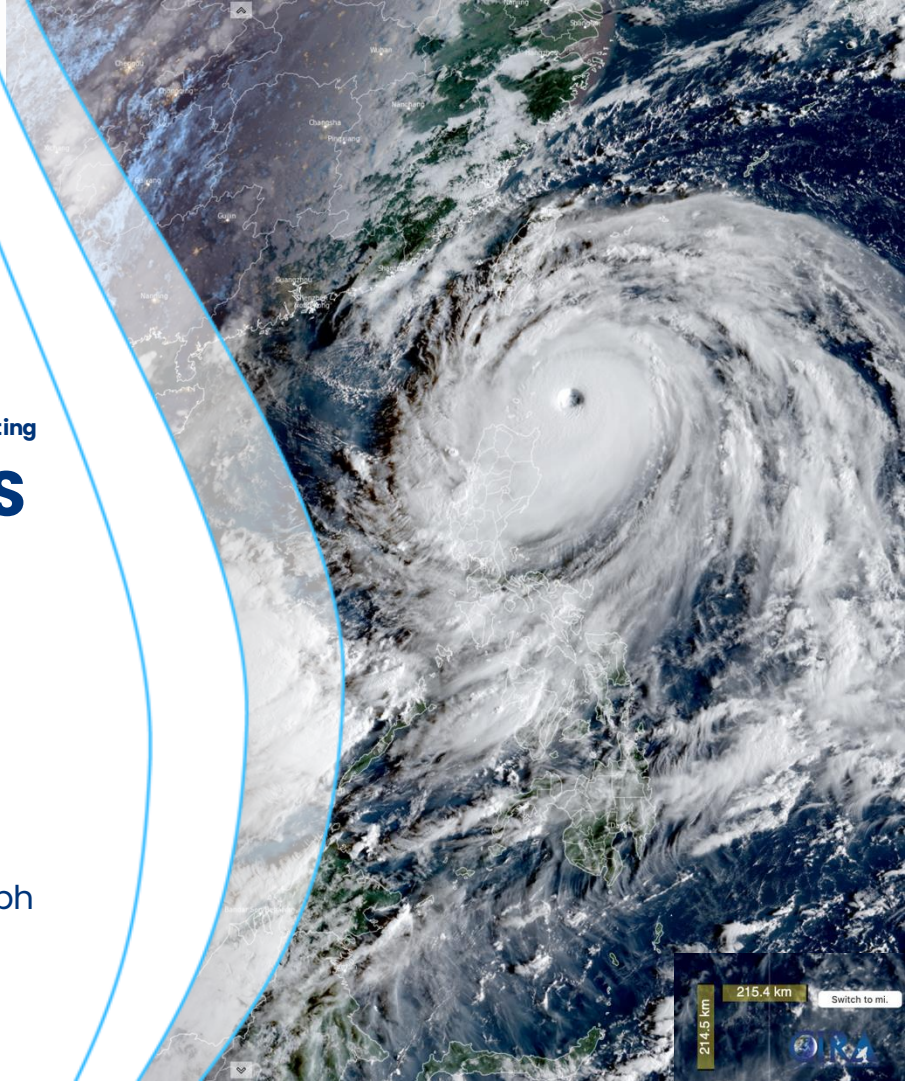
September 22, 2026 2nd Latin American and Caribbean Advances on Wx Forecasting

Novel Global km-scale MPAS Forecasts

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Mesoscale and Microscale Meteorology Laboratory
NSF National Center for Atmospheric Research

Special thanks to Gretchen Mullendore and Everette Joseph

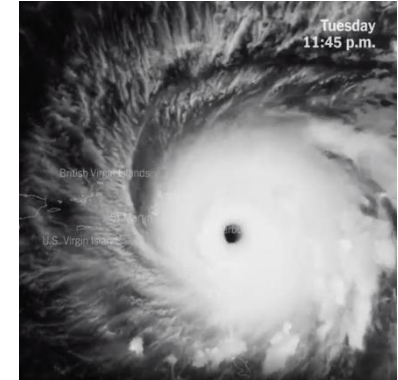
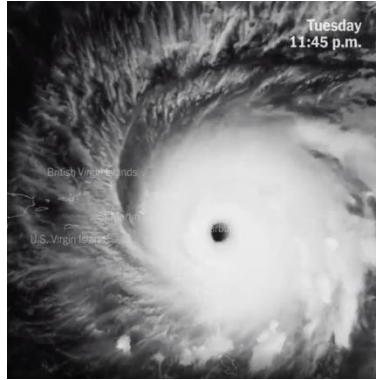
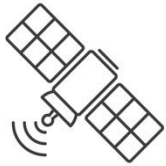
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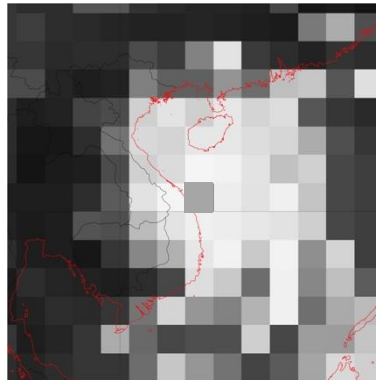
Each step down in grid spacing makes the hurricane more like the satellite image.



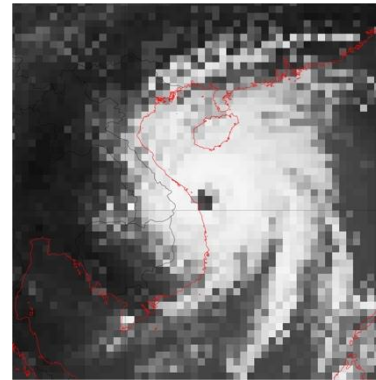
Satellite
image



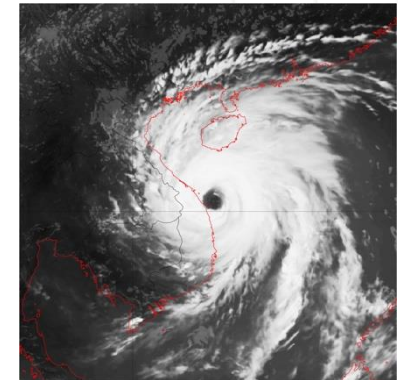
MPAS
Simulation



$\Delta x = 120\text{km}$



$\Delta x = 15\text{ km}$



$\Delta x = 3.75\text{ km}$

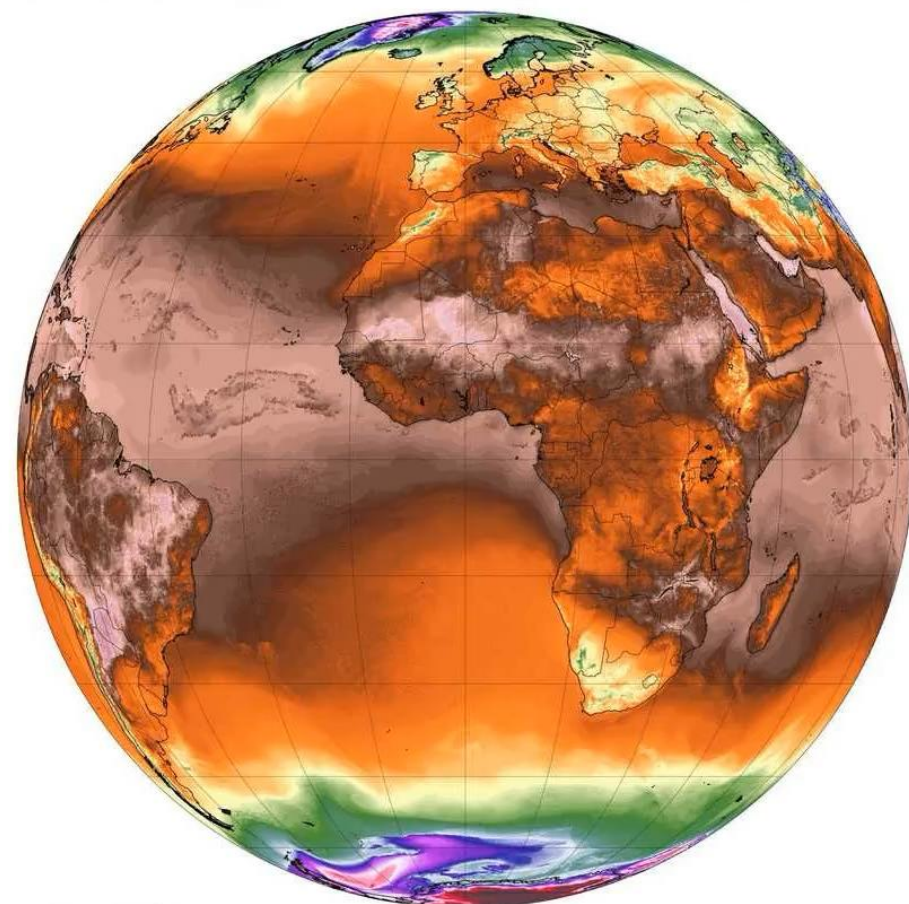
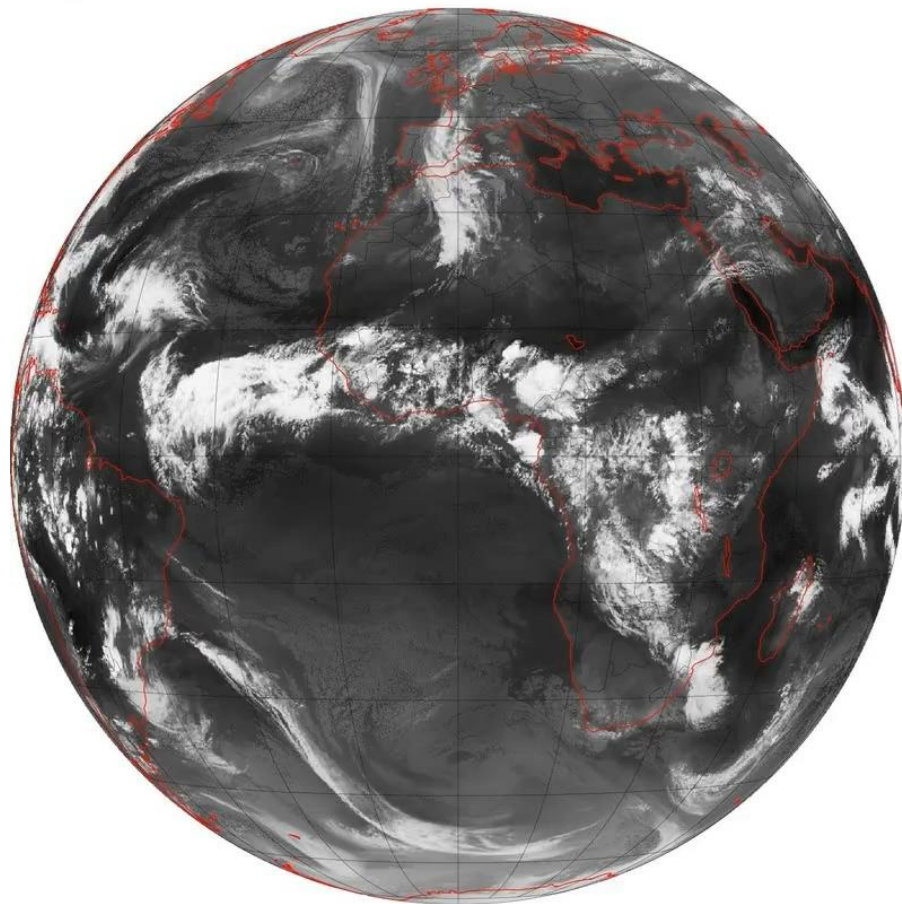
OLR (W/m^2)

2012-10-20_00:00:00

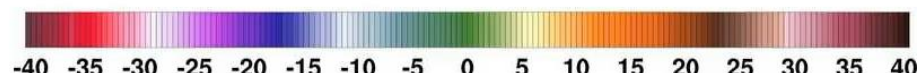
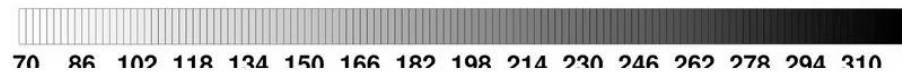
2-m Temperature ($^{\circ}\text{C}$)

4-km MPAS

2012-10-20_00:00:00



Max: 36.5°C
Min: -60.6°C



Forecast setup



	3-km MPAS	3.75-km MPAS
Forecast initialization dates	26 April – 30 May 2025 (35 forecasts)	31 August – 6 October 2025 (37 forecasts)
Forecast length	2.5 days (60 hours)	5 days (120 hours)
Forecast initialization time	00 UTC	
Forecast initialization source	"cold starts" from 0.25° GFS isobaric-level analysis	

Model Specifications



	3-km MPAS	3.75-km MPAS
Model	MPAS v6.1	MPAS v8.0.1
# of columns	65,536,002	41,943,042
Number of vertical levels	55	55
Model top	30 km	30 km
Time step	18 s	22 s
Parameterizations	Microphysics: Thompson Boundary layer: MYNN Land surface: Noah Cumulus: scale-aware Tiedtke Radiation: RRTMG	

Computational Aspects



	3-km MPAS	3.75-km MPAS
HPC system	NSF NCAR Derecho	
Compute resources	19,200 processors (150 nodes) 6% of machine capacity	38,400 processors (300 nodes) 12% of machine capacity
Runtime	~9 hours per forecast	~5 hours per forecast
Disk space per forecast	~563 GB (61 two-dimensional fields per forecast hour, 3-hourly files with 99 variables)	~1.5 TB (77 two-dimensional fields per forecast hour)

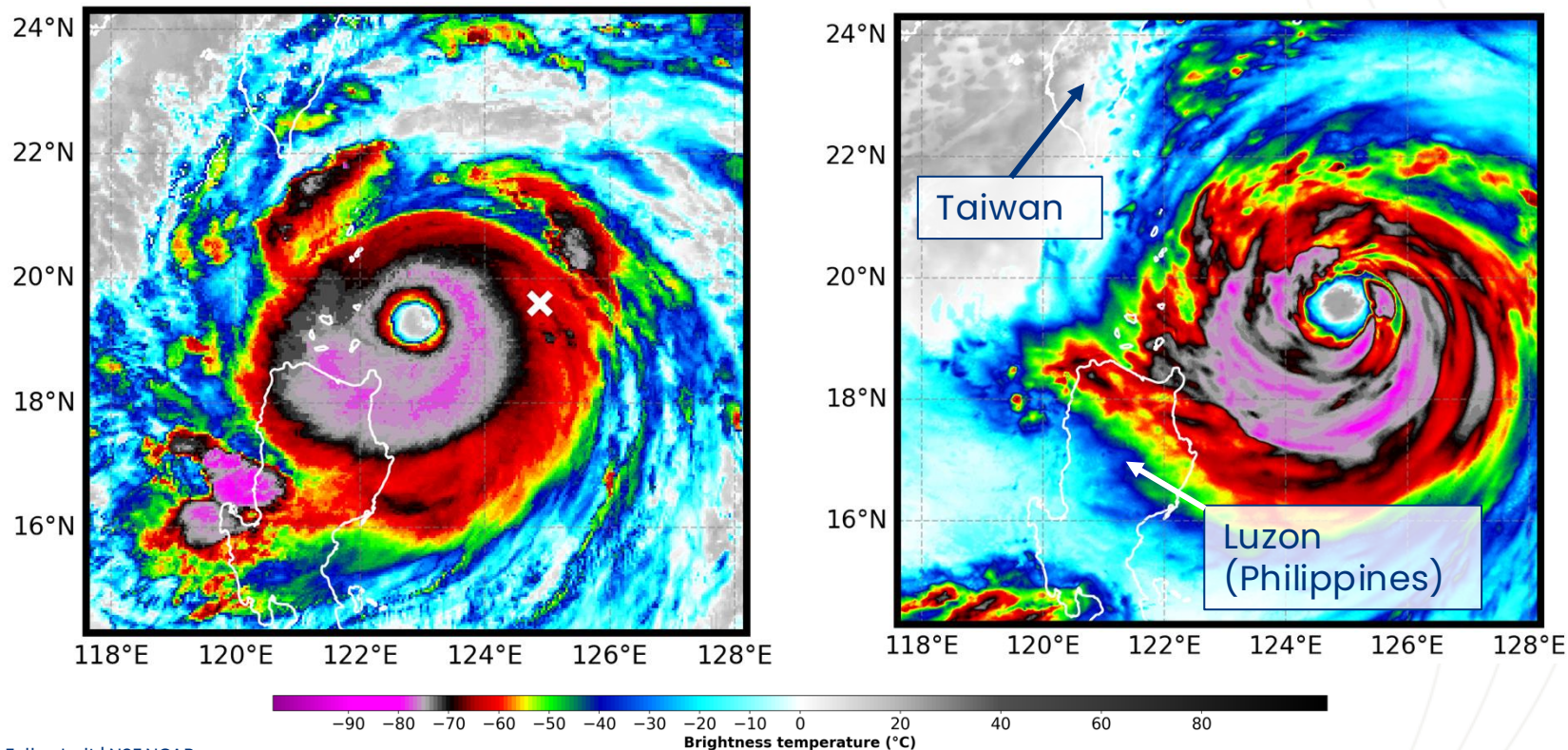
Everything Everywhere All At Once



- One forecast, all over the globe, at the scale of a thunderstorm



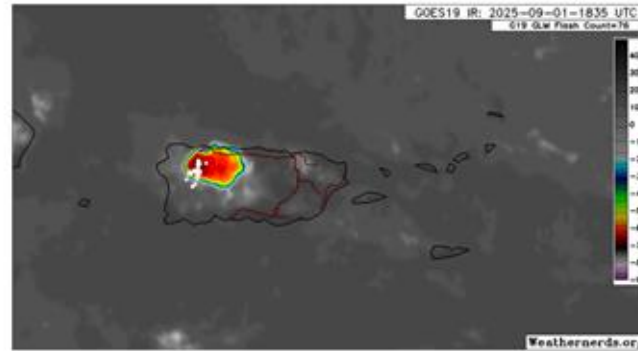
Ragasa: A super typhoon threading Luzon and Taiwan.



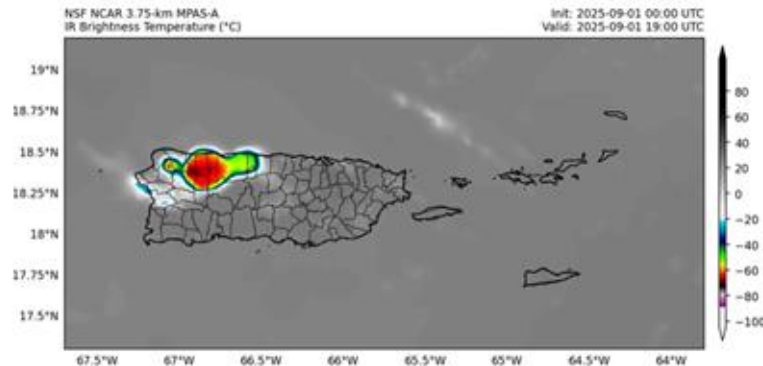
Puerto Rico: A single convective cell— in a global forecast.



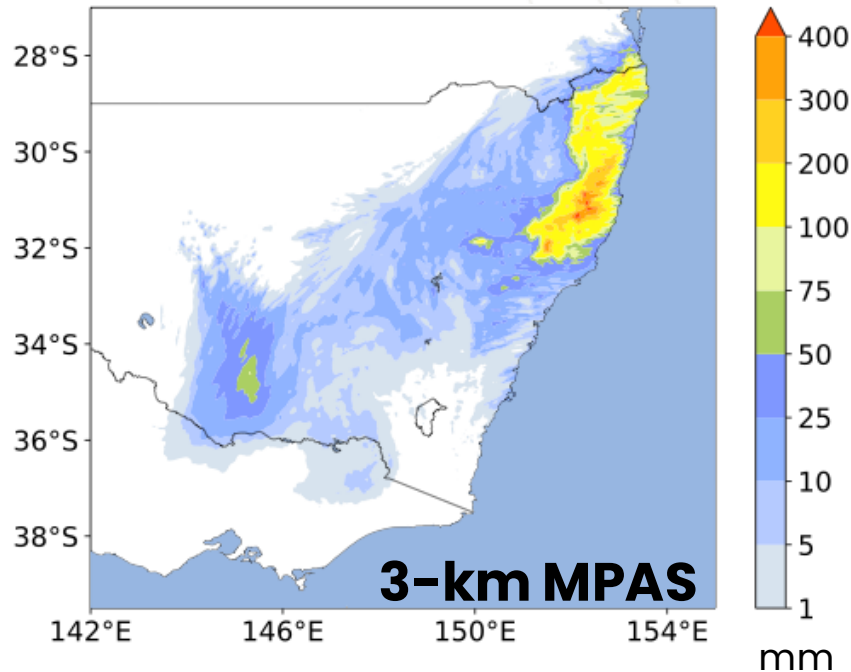
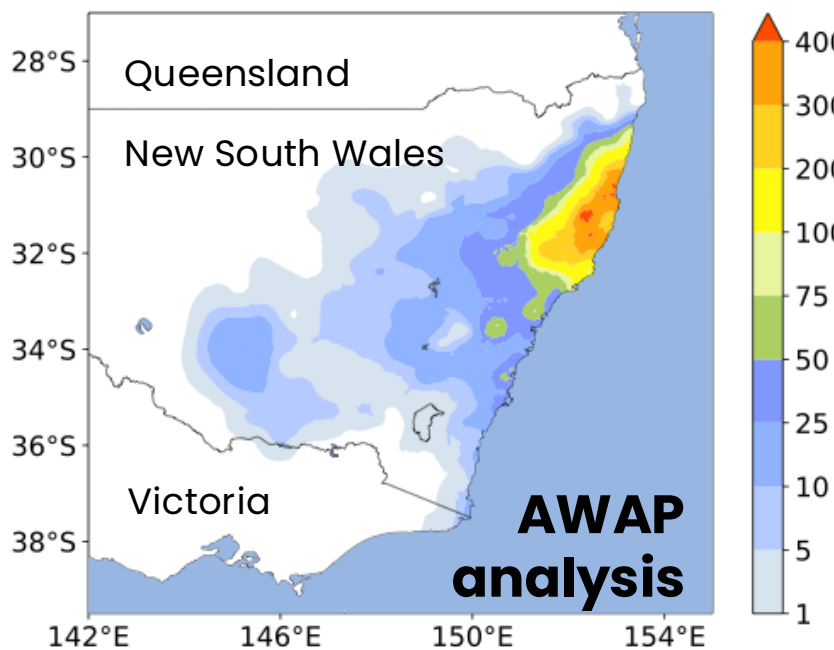
Satellite



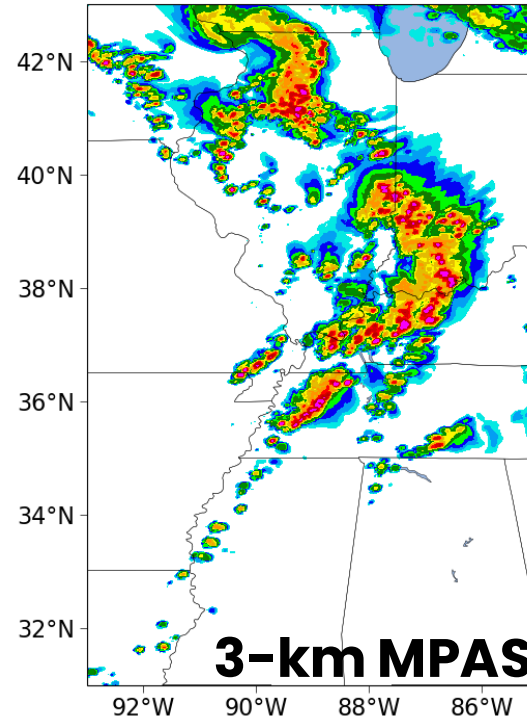
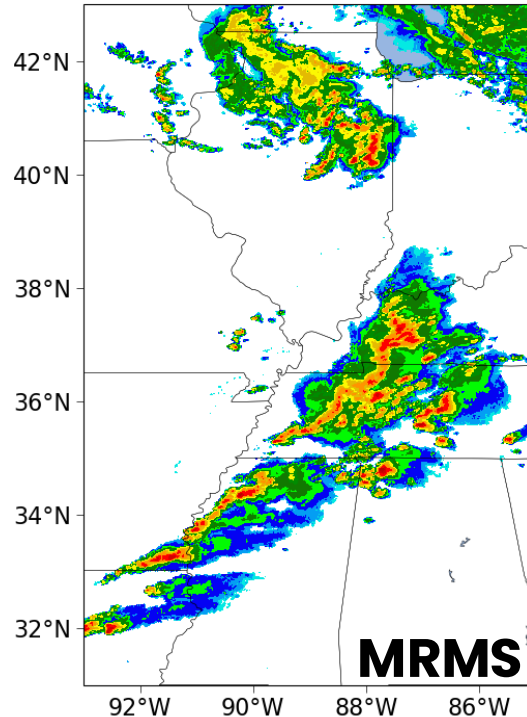
3.75-km MPAS



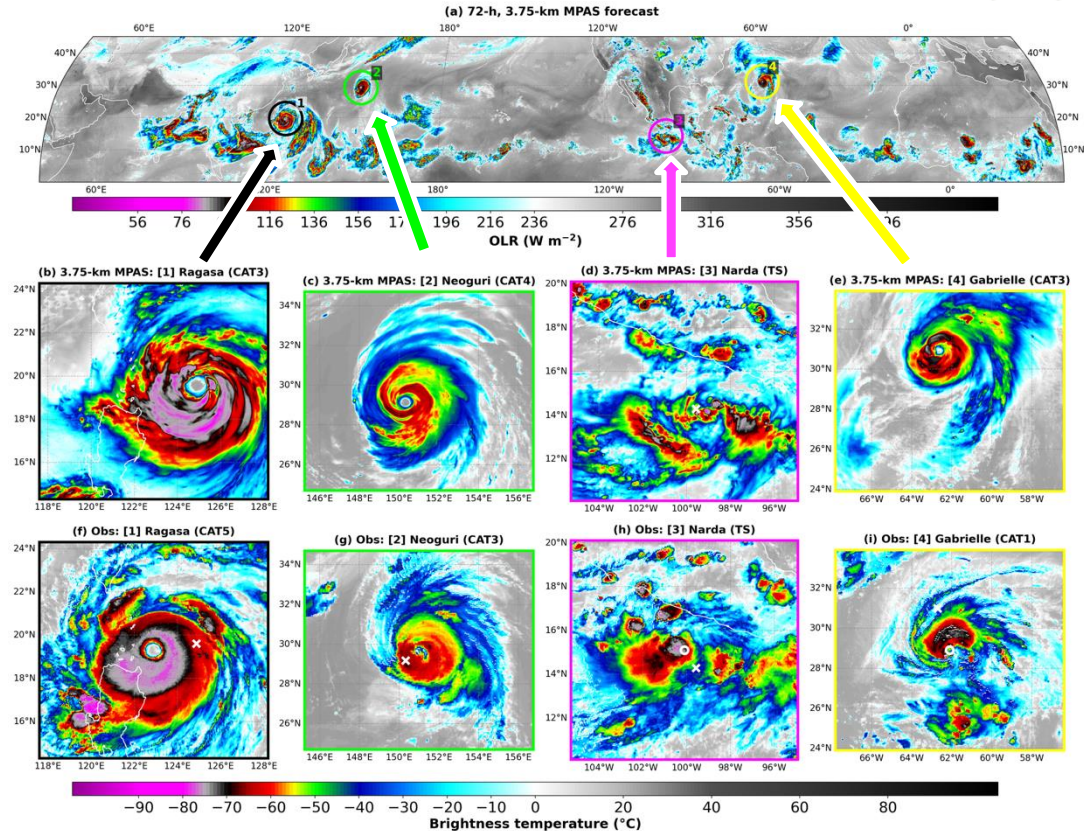
Australia: 48-h accumulated rainfall matches the AWAP analysis across New South Wales.



US severe wx: A 22-h reflectivity forecast with storm mode that MRMS recognizes.



TC field: A 3-day forecast reproduces the planet's whole tropical cyclone population.

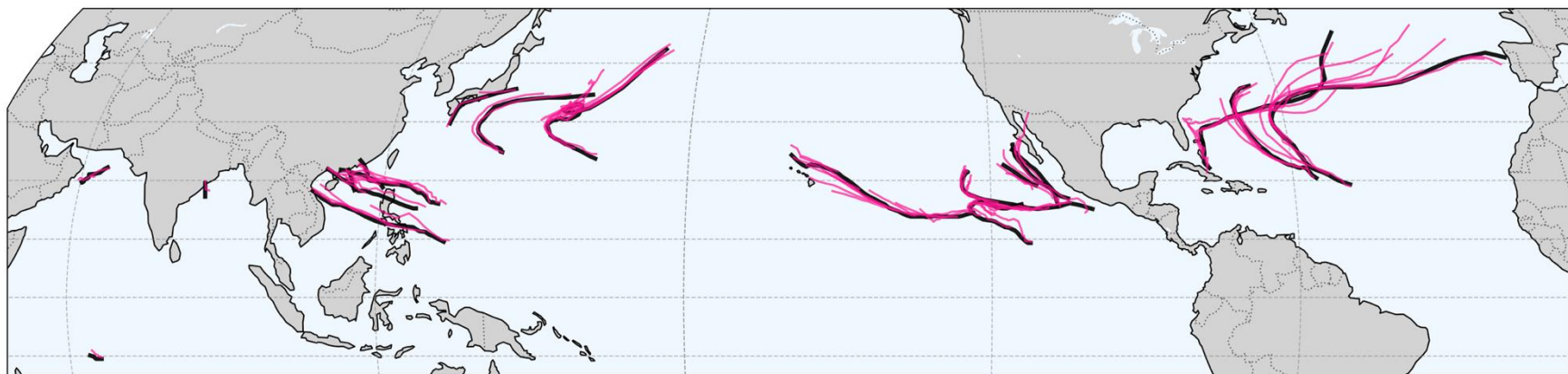


3.75-km MPAS
3-day forecast

Satellite Obs

The 3.75-km runs captured 20 TCs, including six major hurricanes

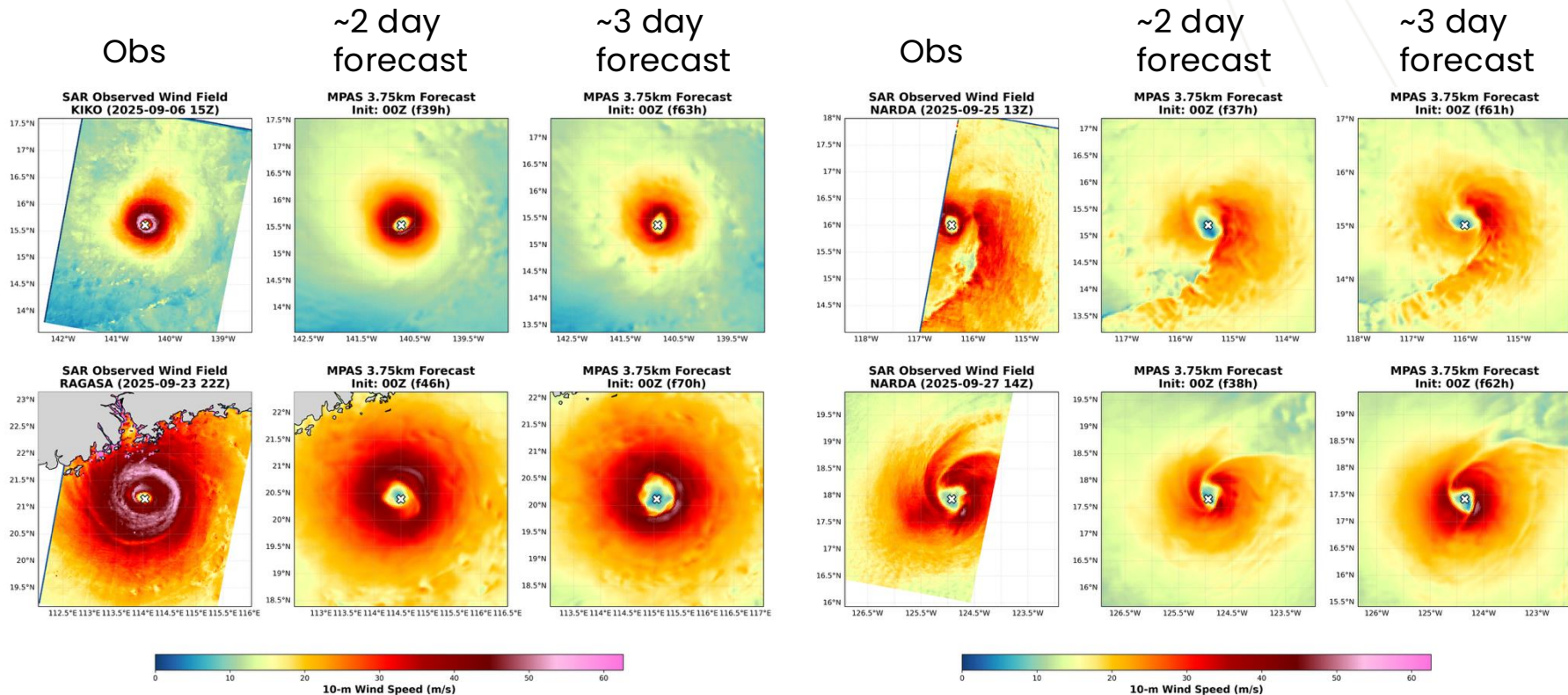
Tropical Cyclone Tracks
August 31 - October 6, 2025



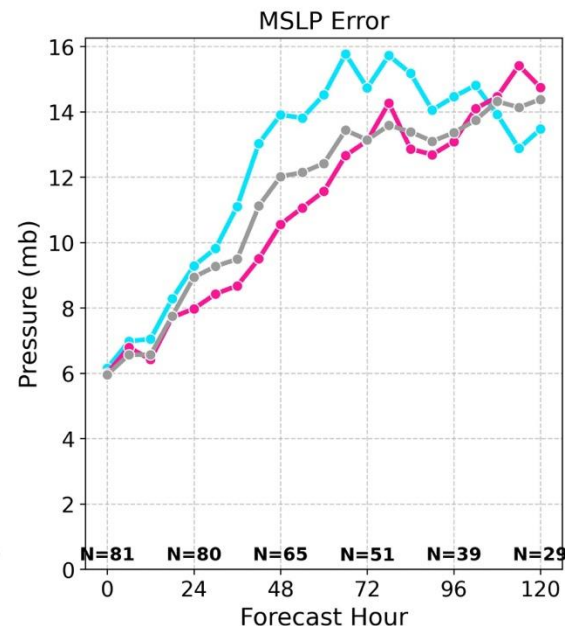
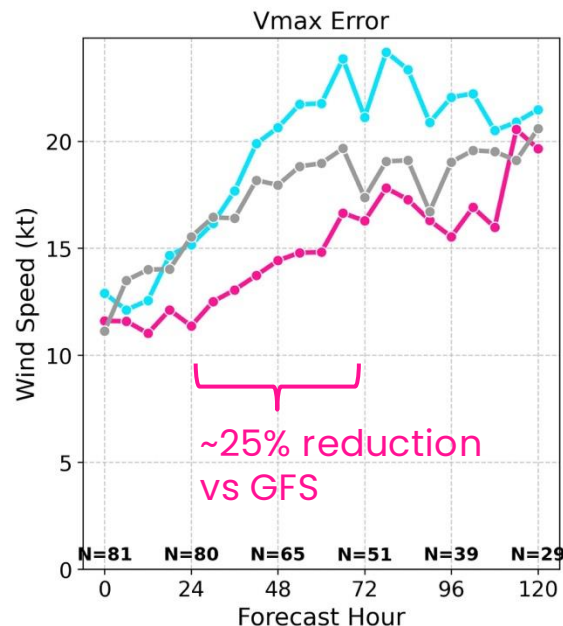
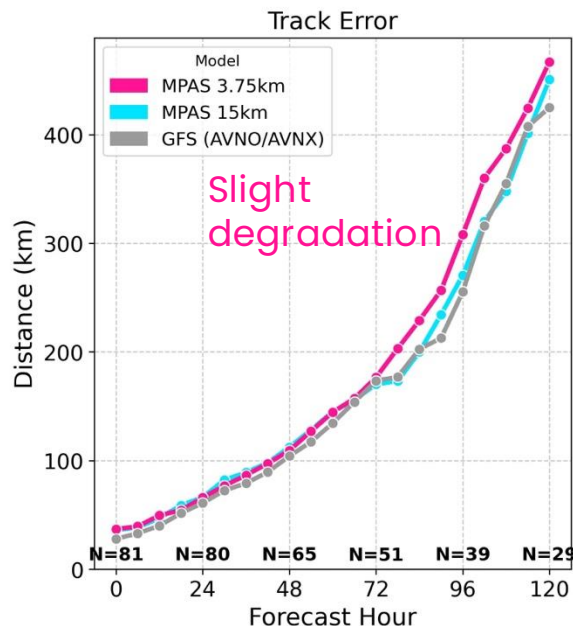
— IBTrACS Observations — MPAS 3.75km Forecasts

TS	5
CAT 1	6
CAT 2	2
CAT 3	1
CAT 4	4
CAT 5	2

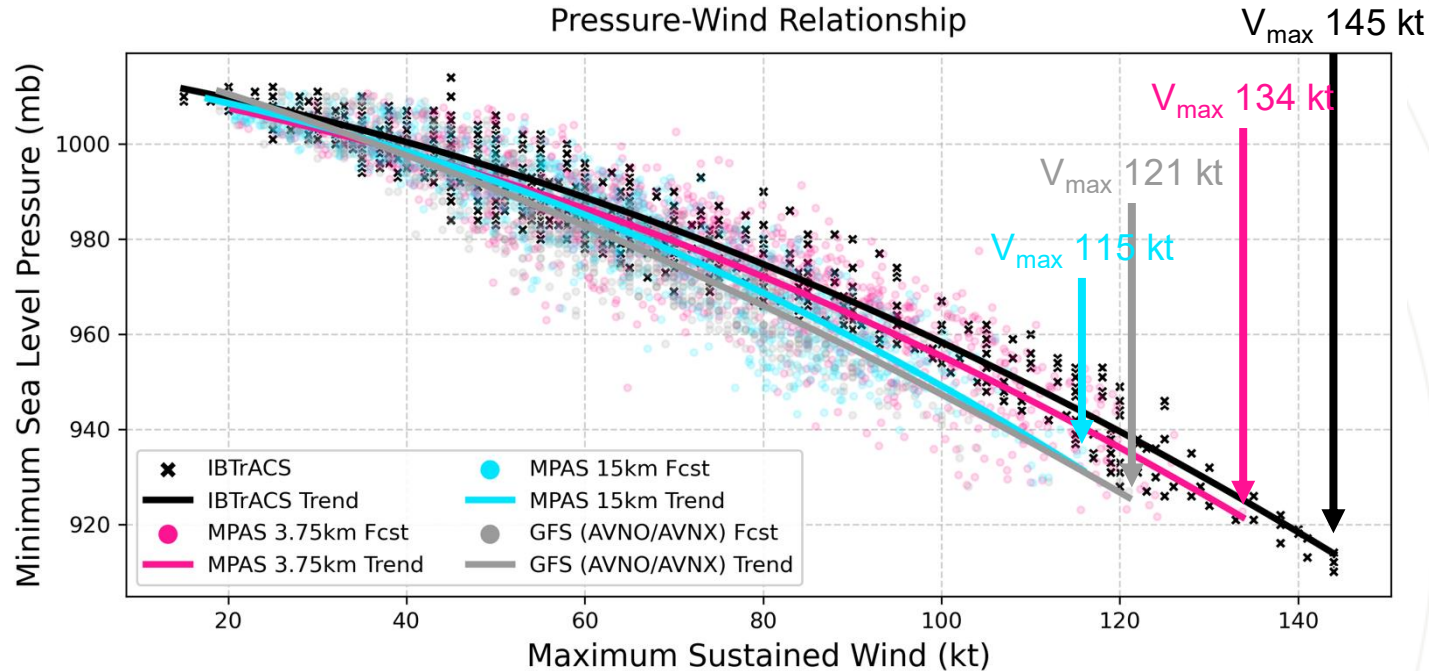
Storm size and structural anomalies are reproduced, not smeared out.



Intensity errors fall ~20–25% against the GFS;
tracks degrade slightly.



The pressure–wind relationship is right; yet Cat-5 winds are still out of reach.



Tight eyewalls are still a challenge: RMW under 25 km stays unresolved.

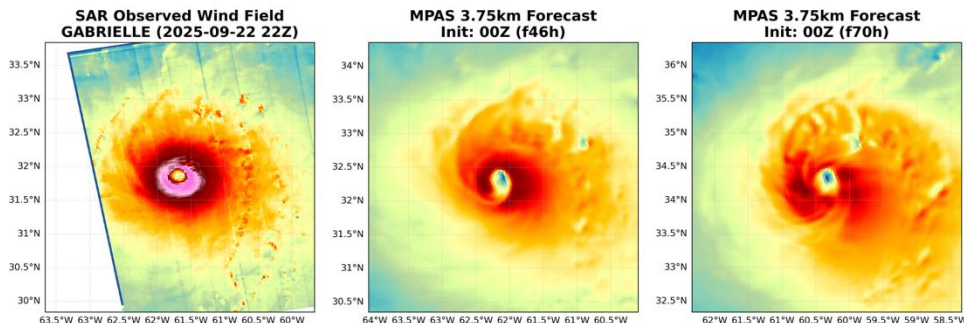


Hurricane Gabrielle

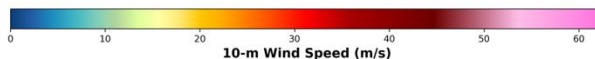
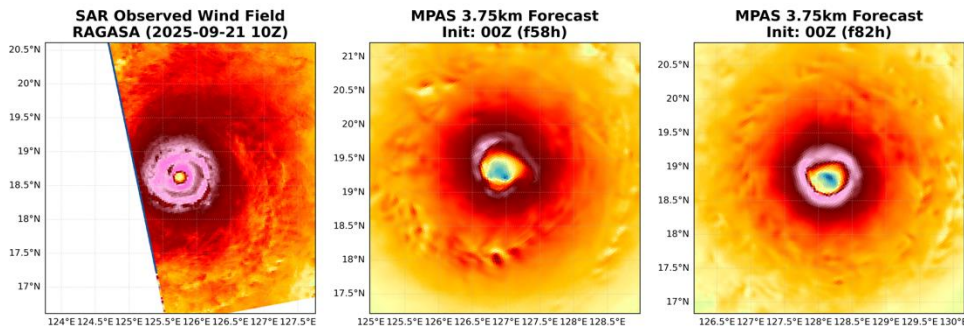
Obs

~2 day
forecast

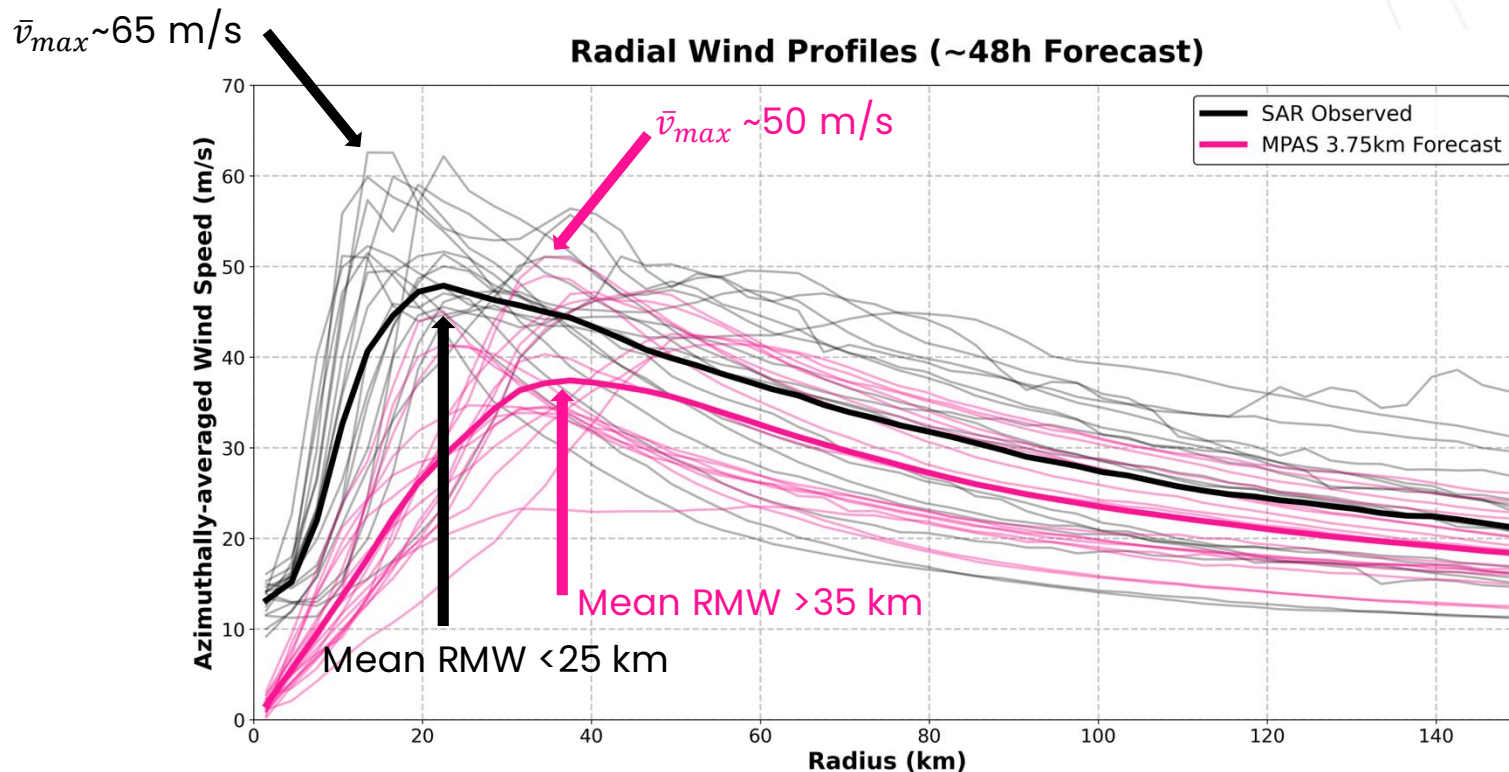
~3 day
forecast



Typhoon Ragasa



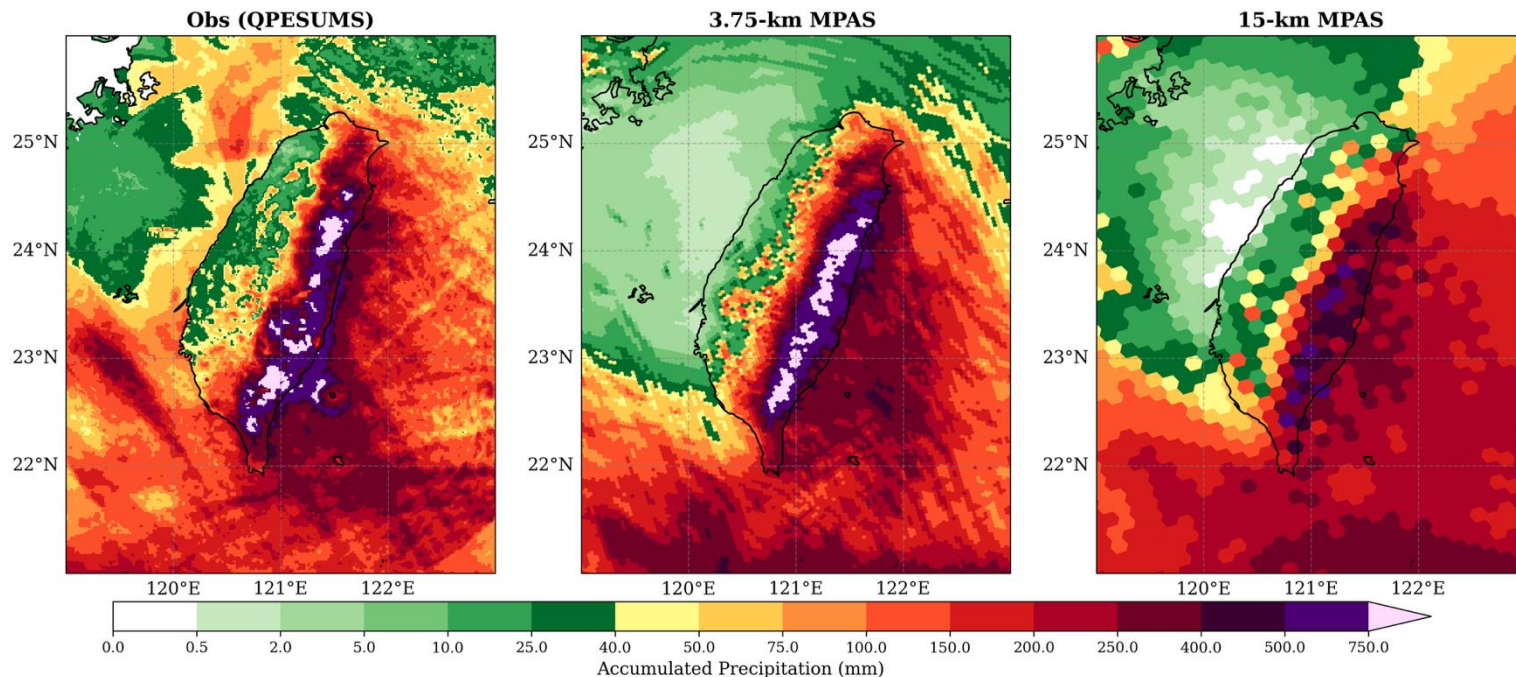
Tight eyewalls are still a challenge: RMW under 25 km stays unresolved.



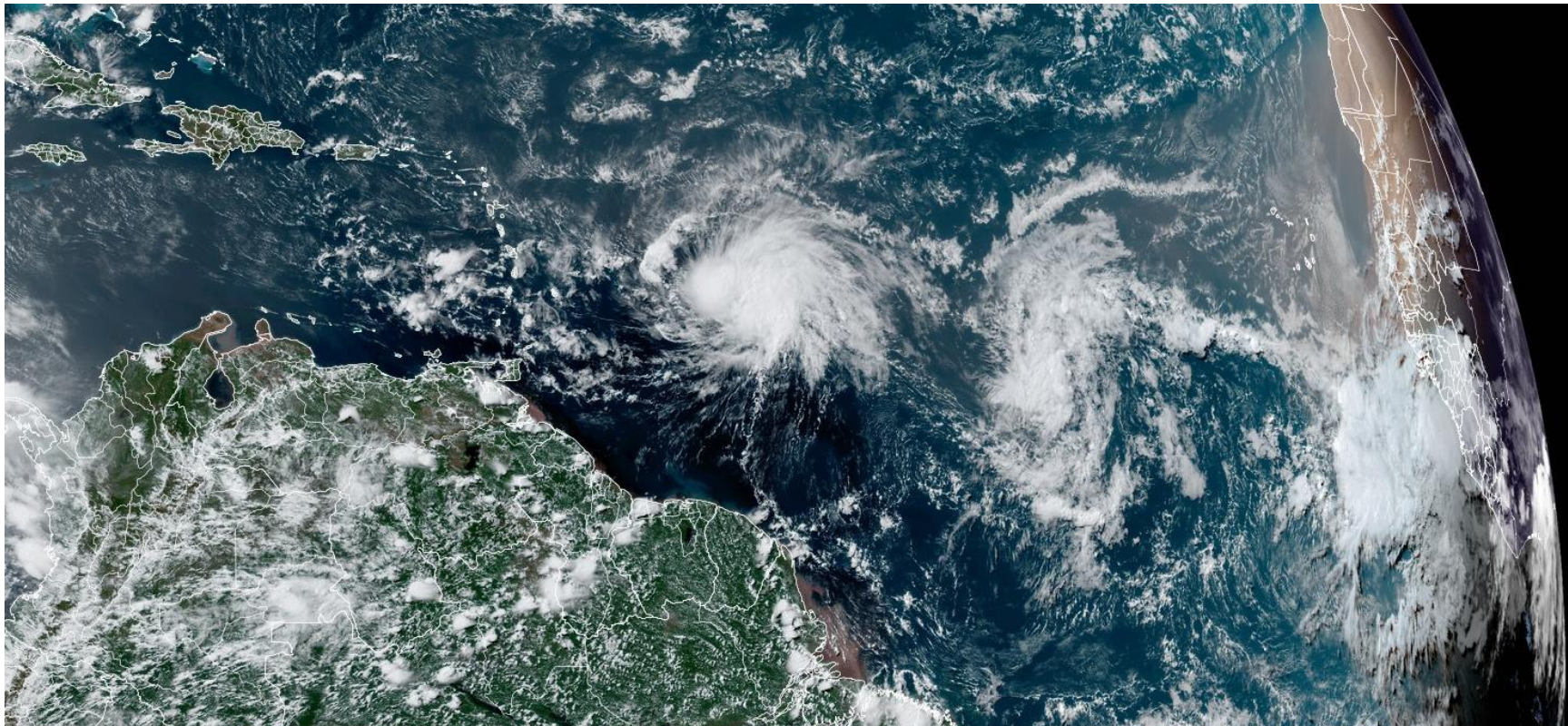
Taiwan's orographic rainfall extreme during Ragasa is resolved.



Typhoon Ragasa (2025) in Taiwan: Accumulated Precipitation Comparison
2025-09-20 00:00 - 2025-09-25 00:00



Tropical waves



Tropical waves stay too weak.



Obs

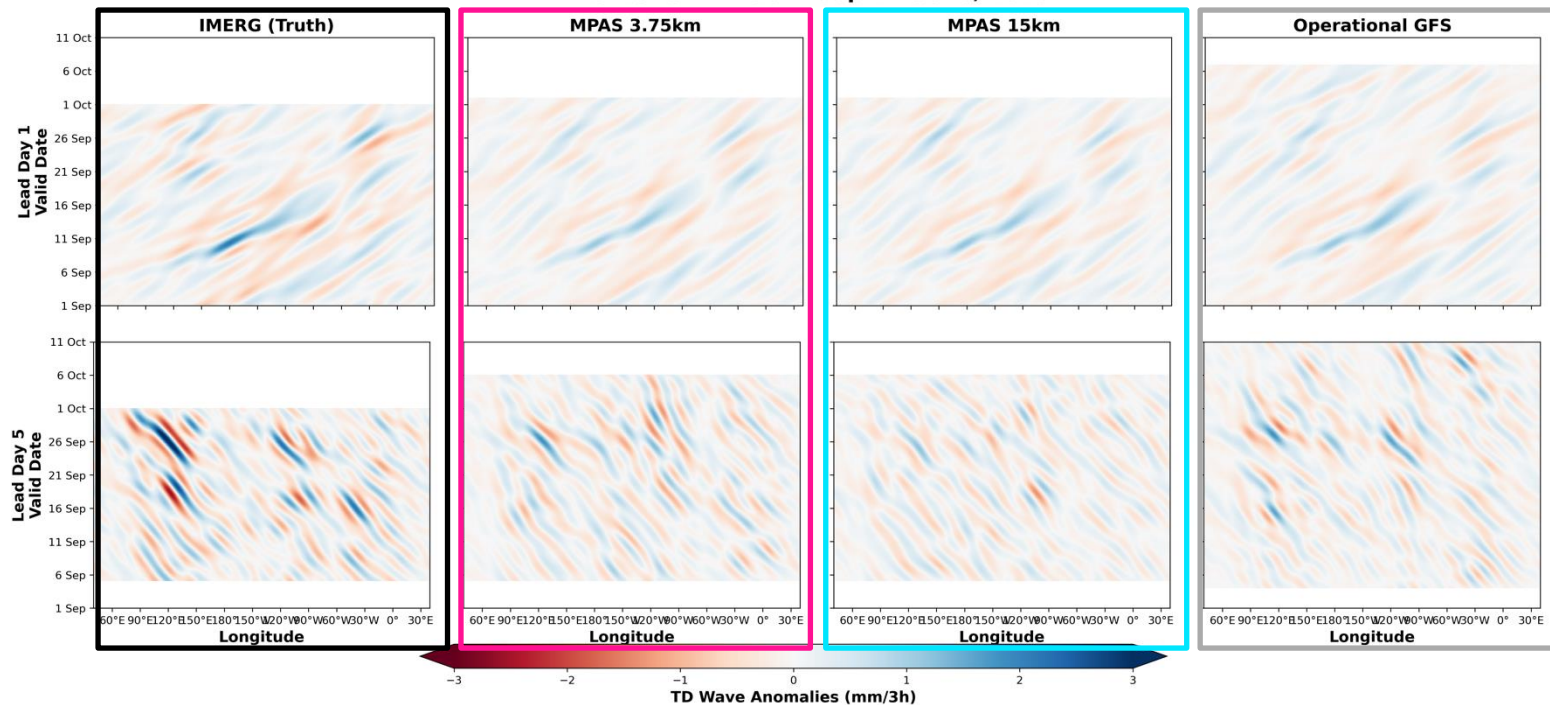
3.75-km MPAS

15-km MPAS

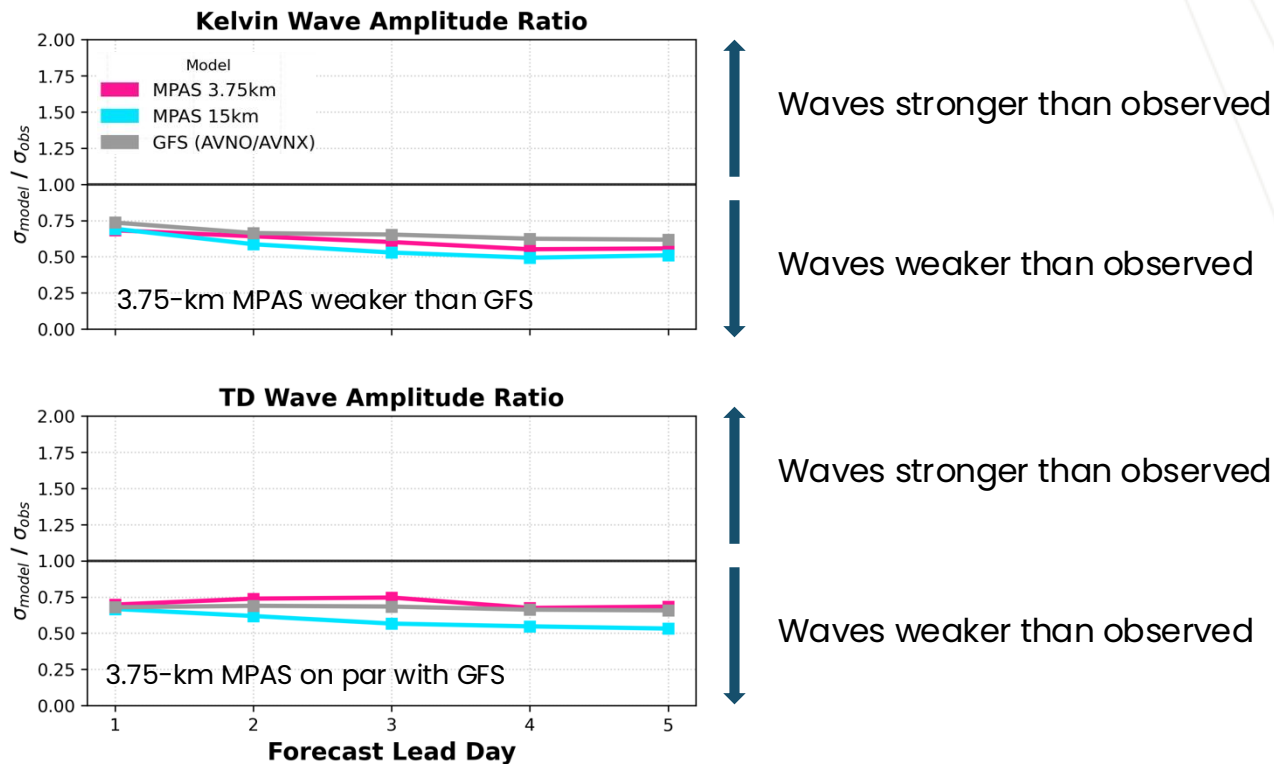
GFS

Kelvin Wave Anomalies & TC Tracks (5°N - 15°N Average)
Initialization Window: Sep 1 - Oct 6, 2025

Kelvin waves
Tropical-depression-type waves

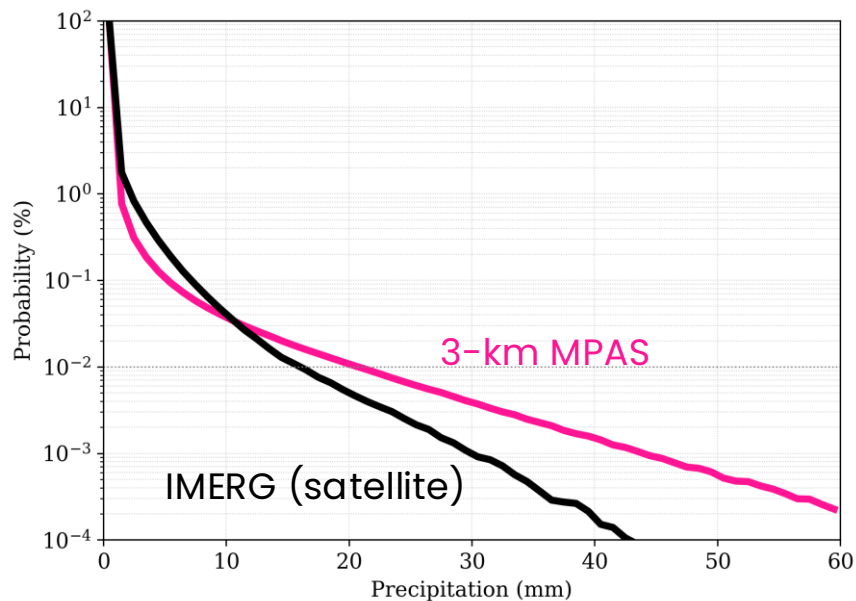


Tropical waves stay too weak.

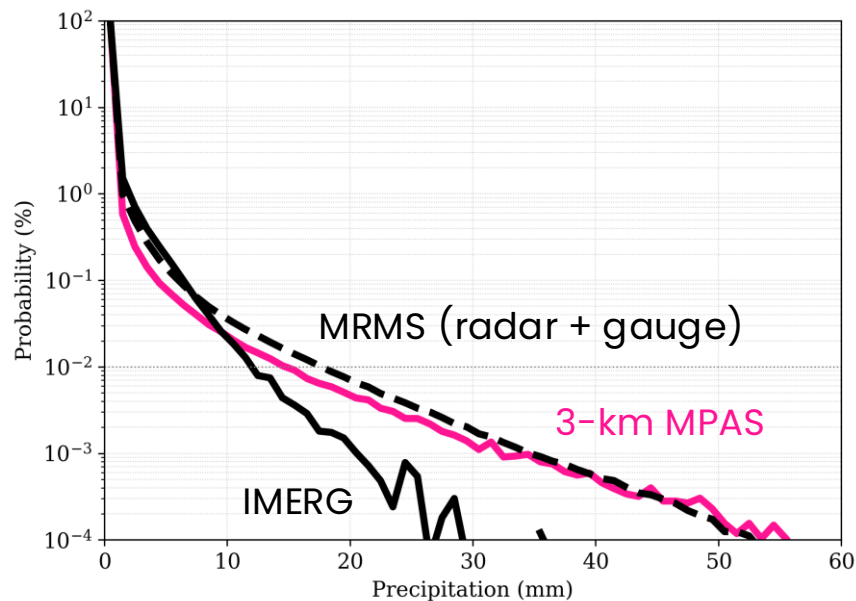


IMERG suggests 3-km MPAS overpredicts heavy rain — but does it really?

Tropics



U.S.



- **Resolution:** eyewalls with RMW < 25 km and secondary eyewall formation need finer than 3.75 km — or continue running additional regional models.
- **Physics:** tropical wave amplitude are not resolution problems alone.
- **Initialization:** every forecast here was a cold start. Coupled, cycled initialization is the obvious next step.
- **Verification:** assessing a global km-scale model is difficult without global km-scale observations.

Take-home points

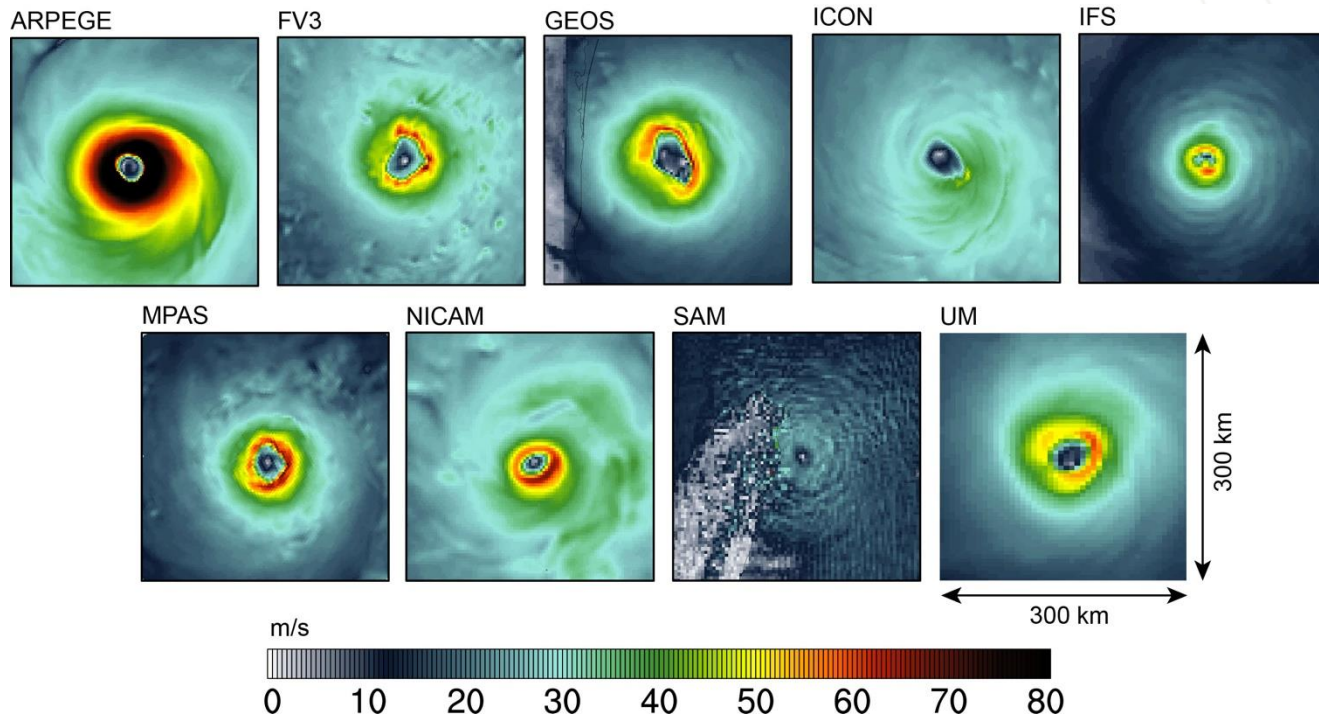


- Global km-scale forecasting is feasible today: 72 real-time forecasts, ~5–9 hours each, on a routine share of NSF NCAR's *Derecho*.
- 3.75-km MPAS beats operational models where it counts: ~20–25% lower TC intensity error than the GFS, and resolved orographic rainfall extremes.
- Structure is still biased — tight eyewalls, concentric eyewalls, and tropical wave amplitude.
- Verification is the binding constraint: no global km-scale observations exist yet.

Extra slides



Global km-scale models exhibit significant variability in hurricane structure and intensity across different models.



Some quantitative numbers

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MPAS 3.75km PERCENTAGE IMPROVEMENT
(Positive % means 3.75km has LOWER error than baseline)

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-- 3.75km Improvement vs. GFS (AVNO/AVNX) --

Tau	Track Impr	Wind Impr	MSLP Impr
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0	-32.1%	-4.2%	-0.8%
6	-19.3%	14.1%	-3.4%
12	-24.0%	21.3%	2.2%
18	-5.2%	13.6%	0.5%
24	-8.2%	26.8%	10.8%
30	-5.9%	24.0%	9.1%
36	-9.4%	20.4%	8.6%
42	-8.7%	24.5%	14.6%
48	-5.0%	19.7%	12.2%
54	-8.3%	21.4%	8.9%
60	-7.5%	21.9%	6.8%
66	-2.1%	15.3%	5.8%
72	-1.8%	6.3%	0.2%
78	-14.8%	6.6%	-5.0%
84	-13.1%	9.6%	3.9%
90	-20.6%	2.5%	3.1%
96	-20.7%	18.4%	2.0%
102	-14.0%	13.6%	-2.5%
108	-9.0%	18.1%	-1.0%
114	-4.1%	-7.6%	-9.1%
120	-9.9%	4.6%	-2.5%

MEAN	-10.5%	13.9%	2.8%
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~20-25% error reduction vs the GFS